Chaining implications under derivability

The Open Logic Project contributors

openlogic:first-order-logic/axiomatic-deduction:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-first-order-logic-axiomatic-deduction/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: in-source stub ex:chain (example environment, not a counted card; its displayed derivation is the author's own and the proof consumes lines 3-7 by explicit reference).

Editorial note. The label prop:chain names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own prop environment in the unit source.

Allowed standard prerequisites (editorial).

- Worked chaining derivation. Same unit, \ollabel{ex:chain} (source line 1561, preceding example). Lines 3-7 of that displayed seven-line derivation are inserted verbatim; the proof checks the renumbering of the Hyp references.

Proposition 0.1. *If* $\Gamma \vdash A \to B$ *and* $\Gamma \vdash B \to C$*, then* $\Gamma \vdash A \to C$

Proof. Suppose $\Gamma \vdash A \to B$ and $\Gamma \vdash B \to C$. Then there is derivation of $A \to B$ from Γ; and derivation of $B \to C$ from Γ as well. Combine these into a single derivation by concatenating them. Now add lines 3–7 of the derivation in the preceding example. This is derivation of $A \to C$—which is the last line of the new derivation—from Γ. Note that the justifications of lines 4 and 7 remain valid if the reference to line number 1 is replaced by reference to the last line of the derivation of $A \to B$, and reference to line number 2 by reference to the last line of the derivation of $B \to C$. □